CS 464 - Introduction to Machine Learning

Homework 1 Report

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1 Software Company Quality Control (15 pts)

1.1 Question 2.1 [5 pts]

Problem: Given that the program contains a fault, find the probability that it was written by Team A or B.

Step 1: Define Given Data

- Team A: 50 programmers, fault probability per programmer: $P(F|A) = \frac{2}{100}$
- Team B: 30 programmers, fault probability per programmer: $P(F|B) = \frac{5}{100}$
- Team C: 20 programmers, fault probability per programmer: $P(F|C) = \frac{10}{100}$
- Probability of selecting a team:

$$P(A) = \frac{50}{100} = \frac{1}{2},$$
$$P(B) = \frac{30}{100} = \frac{3}{10},$$
$$P(C) = \frac{20}{100} = \frac{1}{5}$$

Step 2: Compute Total Probability of a Faulty Program Using the law of total probability:

$$P(F) = P(F|A)P(A) + P(F|B)P(B) + P(F|C)P(C) \tag{1}$$

Substituting values:

$$P(F) = \left(\frac{2}{100} \times \frac{1}{2}\right) + \left(\frac{5}{100} \times \frac{3}{10}\right) + \left(\frac{10}{100} \times \frac{1}{5}\right) = \frac{9}{200} \tag{2}$$

Step 3: Compute $P(A \cup B|F)$ using Bayes' Theorem

$$P(A|F) = \frac{P(F|A)P(A)}{P(F)} = \frac{\left(\frac{2}{100} \times \frac{1}{2}\right)}{\frac{9}{200}} = \frac{2}{9} \tag{3}$$

$$P(B|F) = \frac{P(F|B)P(B)}{P(F)} = \frac{\left(\frac{5}{100} \times \frac{3}{10}\right)}{\frac{9}{200}} = \frac{1}{3} \tag{4}$$

$$P(A \cup B|F) = P(A|F) + P(B|F) = \frac{5}{18} + \frac{1}{3} = \frac{5}{9} \tag{5}$$

1.2 Question 2.2 [5 pts]

Problem: Given that the program was not written by Team A, find the probability that it was written by Team C.

Solution: Using conditional probability:

$$P(C|\text{not } A) = \frac{P(C)}{P(B) + P(C)} = \frac{\frac{1}{5}}{\frac{3}{10} + \frac{1}{5}} = \frac{2}{5} \tag{6}$$

1.3 Question 2.3 [5 pts]

Problem: Given that the program is not faulty, find the probability that it was written by Team A.

Solution:

$$P(\text{not } F) = \left(\frac{98}{100} \times \frac{1}{2}\right) + \left(\frac{95}{100} \times \frac{3}{10}\right) + \left(\frac{90}{100} \times \frac{1}{5}\right) = \frac{191}{200} \tag{7}$$

$$P(A|\text{not } F) = \frac{\left(\frac{98}{100} \times \frac{1}{2}\right)}{\frac{191}{200}} = \frac{98}{191} \tag{8}$$

2 A Robot's Optimal Score (10 pts)

2.1 Question 3.1 [10 pts]

Problem: Compute the expected score for the robot for $p = 0.2$ and $p = 0.1$.

Solution to Question 3.1

To solve this problem optimally, let's define the expected score S when the robot follows the best strategy.

Step 1: Define the Expected Score Equation

- If the robot presses the **red button**, it gets **8 points** immediately.
- If the robot presses the **blue button**, it gets **10 points with probability** p, and with probability $1 - p$, it gets **0 points** and must choose again.

Thus, the expected score S when choosing the blue button is given by:

$$S = p \cdot 10 + (1 - p) \cdot S \tag{9}$$

Rearrange the equation:

$$\begin{aligned} S - (1-p)S &= 10p \\ S(1 - (1-p)) &= 10p \\ pS &= 10p \\ S &= \frac{10p}{p} = \frac{10p}{1 - (1-p)} = 10 \end{aligned}$$

Step 2: Compare with the Red Button's Score

The robot will choose the blue button if:

$$S > 8 \tag{10}$$

For $p = 0.2$:

$$S = \frac{10 \times 0.2}{1 - 0.8} = \frac{2}{0.2} = 10$$

Since $10 > 8$, the robot should choose the blue button, and the score is **10**.
For $p = 0.1$:

$$S = \frac{10 \times 0.1}{1 - 0.9} = \frac{1}{0.1} = 10$$

Since $10 > 8$, the robot should again choose the blue button, and the score is **10**.

Final Answer

For both $p = 0.2$ and $p = 0.1$, the optimal strategy is to press the blue button, giving an average score of **10**.

3 Stellar Classification (75 pts)

3.1 Question 4.1 [25 pts]

Problem: Train a Naïve Bayes model on the training set and evaluate your model on the test set given.
Solution:

- Model Accuracy: **0.8788**
- Confusion Matrix:

$$\begin{bmatrix} 5703 & 1229 \\ 848 & 9363 \end{bmatrix} \tag{11}$$

3.2 Question 4.2 [5 pts]

Problem: Explain why we do not estimate the joint probability of the features while training the Naïve Bayes classifier.
Solution:

- Estimating joint probabilities requires a massive dataset to cover all possible feature combinations.
- Many feature combinations may never appear in training, leading to zero probabilities.
- Naïve Bayes assumes feature independence to make calculations feasible and avoid overfitting.

3.3 Question 4.3 [5 pts]

Problem: Report the number of parameters that need to be estimated in this model. Show your work.

Solution:

- In Naïve Bayes, we estimate $P(\text{feature}|\text{class})$ for all possible feature values. Additionally, we have 1 parameter for the prior probability $P(Y = \text{Galaxy})$.
- The total number of parameters is computed as:

$$1 + \sum_{i=1}^{9} (\#\text{values of feature } i \times 2 \text{ (classes)})$$

$$= 1 + 2 \times (6 + 6 + 6 + 13 + 9 + 33 + 33 + 8 + 2)$$

$$= \mathbf{233}$$

- In reality, the number of values per feature may vary, and if features are continuous, we estimate mean and variance instead of discrete probabilities.

3.4 Question 4.4 [5 pts]

Problem: Report the top 3 feature values that lead to classifying an object as a galaxy and the top 3 feature values that lead to classifying an object as otherwise.

Solution:

Table 1: **Top 3 Feature Values Indicating a Galaxy**

Feature Value	**Feature**	**Score**
Low	Redshift	16.003
Medium	Redshift	10.358
U_27	Ultraviolet Filter	8.797

Table 2: **Top 3 Feature Values Indicating NOT a Galaxy**

Feature Value	**Feature**	**Score**
Very High	Redshift	0.0007
Sagittarius	Alpha	0.0170
Taurus	Alpha	0.0960

- Low Redshift is the strongest indicator of a Galaxy, contributing a score of 16.003. U27 Ultraviolet Filter is also a strong indicator, meaning that galaxies often exhibit higher ultraviolet emissions. Very High Redshift is the strongest indicator of a Non-Galaxy, meaning such objects are likely farther away or different in nature.

3.5 Question 4.5 [10 pts]

Problem: Rank features based on their mutual information with the class variable and report it as a table.

Solution:

- Redshift has the highest MI score (0.18899), meaning it contributes the most information about whether an object is a Galaxy.
- Infrared and Ultraviolet filters are also highly informative, which aligns with astronomical observations that galaxies emit strongly in these wavelengths.
- from figure 1, most precise result is given by k=4 except k=1 which is not trustable.

Table 3: Feature Ranking Based on Mutual Information and Top Contributing Values

Rank	Feature	MI Score	Top Contributing Value
1	Redshift	0.188998	2.0
2	Infrared Filter	0.061233	10.0
3	Ultraviolet Filter	0.060023	11.0
4	Green Filter	0.059723	13.0
5	Near Infrared	0.033331	9.0
6	Alpha	0.010692	2.0
7	Red Filter	0.012991	10.0
8	Delta	0.002703	1.0
9	Cosmic Ray Activity	0.000007	1.0

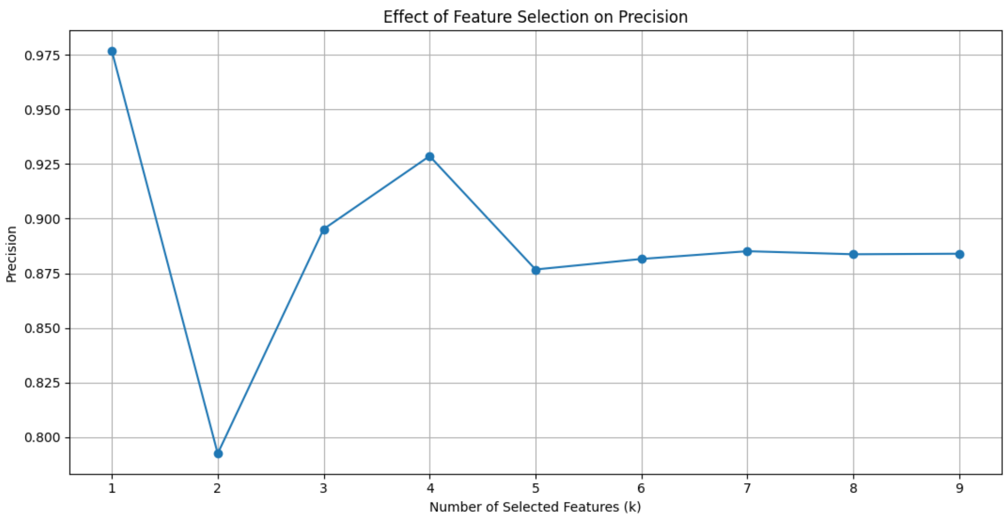

Figure 1: k Values

3.6 Question 4.6 [15 pts]

Problem: Train a hybrid Naïve Bayes classifier where you do not assume that 4 features (redshift, infrared filter, alpha, and delta) are conditionally independent given the class variable. Compare accuracy with your result in Question 4.1.

Solution:

- **Naïve Bayes Accuracy:** 0.8788
- **Hybrid Naïve Bayes Accuracy:** 0.9140
- The Hybrid model captures dependencies between Redshift, Infrared, Alpha, and Delta, leading to better accuracy.
- In the original Naïve Bayes model, all features were treated as independent given the class.
- However, in real-world astronomy, Redshift, Infrared Emission, and Spectral Alpha Angle are highly correlated because galaxies with high redshift often have strong infrared emissions.
- By relaxing the independence assumption for these features, the Hybrid Naïve Bayes model better captures these natural dependencies, leading to improved accuracy.

3.7 Question 4.7 [10 pts]

Problem: Imagine a device that generates an infinite amount of data where the features are conditionally independent with the class variable. If you train your Naïve Bayes model using this dataset and then test it on the same data (training), would you expect the model to achieve 100% accuracy? Justify your response.

Solution:

- Even though features are conditionally independent in this hypothetical dataset, achieving 100 percent accuracy is unlikely due to:
  - Floating-point precision errors, which cause small numerical inconsistencies.
  - Unseen data variations, where certain test cases might have slightly different distributions.
  - Real-world noise, as even clean datasets contain slight misclassifications.

4 OUTPUT

Output is added below and output related k function is above as plot

```
Model Accuracy: 0.8788426763110307
Confusion Matrix:
 [[5703 1229]
 [ 848 9363]]
Hybrid Model Accuracy: 0.9136090532578895
Redshift MI Score: 0.188998, Top Contributing Value: 2.0
Alpha MI Score: 0.010692, Top Contributing Value: 2.0
Delta MI Score: 0.002703, Top Contributing Value: 1.0
Green MI Score: 0.059723, Top Contributing Value: 13.0
NearIR MI Score: 0.033331, Top Contributing Value: 9.0
Cosmic MI Score: 0.000007, Top Contributing Value: 1.0
Red MI Score: 0.012991, Top Contributing Value: 10.0
UV MI Score: 0.060023, Top Contributing Value: 11.0
IR MI Score: 0.061233, Top Contributing Value: 10.0
Top 3 Important Features: ['Redshift', 'IR', 'UV']
Top 3 Important Feature Values: [2.0, 10.0, 11.0]
Your Top 3 Feature Values Indicating NOT a Galaxy:
 Very High redshift     4.0
Sagittarius alpha      11.0
Taurus alpha           10.0
dtype: float64
Your Top 3 Feature Values Indicating a Galaxy:
 Low redshift                2.0
Medium redshift            10.0
U_27 ultraviolet_filter    11.0
dtype: float64
```

Figure 2: output